

OEM TECHNICAL MANUAL

Centrifugal Pump Operations & Maintenance

1. General Information

This technical manual outlines the specifications, operating parameters, and maintenance guidelines for the industrial centrifugal pump series. The instructions within are specific to the unit tagged as **Pump-101**.

The equipment is designed for continuous duty industrial fluid transfer under specified conditions. Deviation from these parameters may result in premature component failure and void the warranty.

--- Page 1 ---

2. Technical Specifications

The centrifugal pump requires strict adherence to operating parameters to ensure longevity and safety. Below are the core technical constraints for Pump-101:

- **Rated operating pressure:** 12 bar
- **Maximum operating temperature:** 150°C
- **Flow capacity:** 450 m³/hr
- **Seal type:** mechanical seal, single cartridge design

3. Component List

The primary sub-assemblies and components of Pump-101 include, but are not limited to:

- Impeller
- Mechanical Seal (Single Cartridge)
- Bearing Housing
- Coupling

--- Page 2 ---

4. Maintenance Guidelines

Routine maintenance is critical for the reliable operation of the pump. The OEM recommends a preventative maintenance schedule to minimize unplanned downtime.

Critical Maintenance Interval: The OEM-recommended seal replacement interval is strictly **every 12 months**, regardless of operational hours, to prevent catastrophic fluid leakage.

Lubrication of bearings should occur every 3,000 operational hours or every 6 months, whichever comes first. Always use OEM-certified replacement parts for internal components.

--- Page 3 ---

5. Safety Precautions

Strict safety protocols must be observed before initiating any inspection, diagnostic, or maintenance activity on Pump-101.

- **Lockout-Tagout (LOTO):** A rigorous lockout-tagout procedure must be implemented on the motor's electrical supply and all associated system valves prior to maintenance.
- **System Depressurization:** Ensure the process line is fully depressurized and drained of all hazardous fluids before loosening any casing bolts or pipe connections.
- **Proper PPE:** Operators and maintenance personnel must wear appropriate Personal Protective Equipment, including safety goggles, heavy-duty gloves, and steel-toed footwear during all manual interventions.

--- Page 4 ---